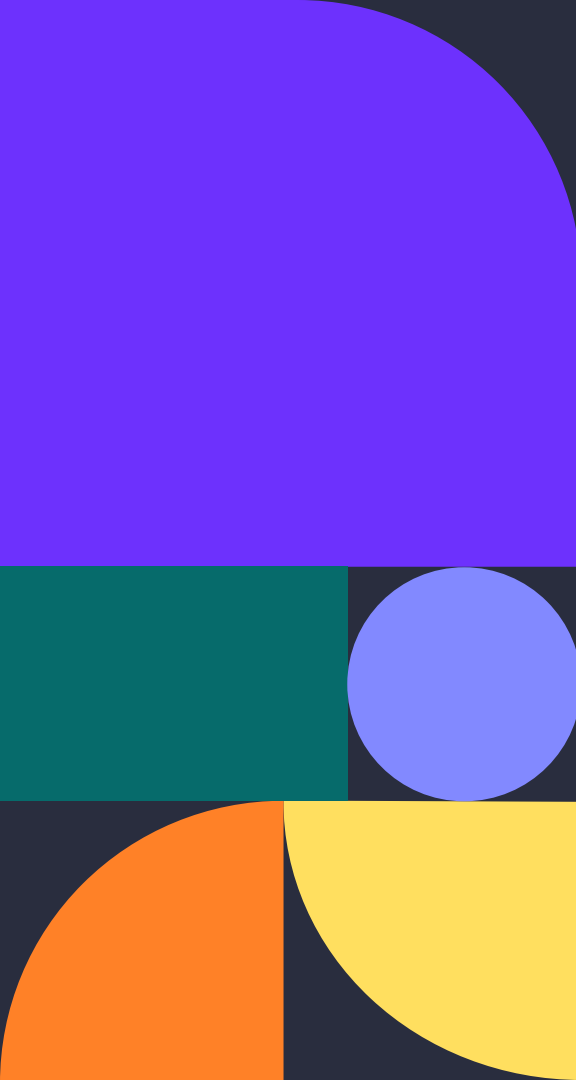




Text-Guided Video Object Erasure Agent

Group 12

Project Members

- Rishab K S
251110061
- Kanchi Kandadai Bharath
251110042
- Hemant Jagtap
251110037
- Ashwin Kansagara
251110019
- Abhishek Dixit
251110003
- Nimmala Rithika Reddy
220716
- Sandela Sai Amruta Varshini
220960

Abstract

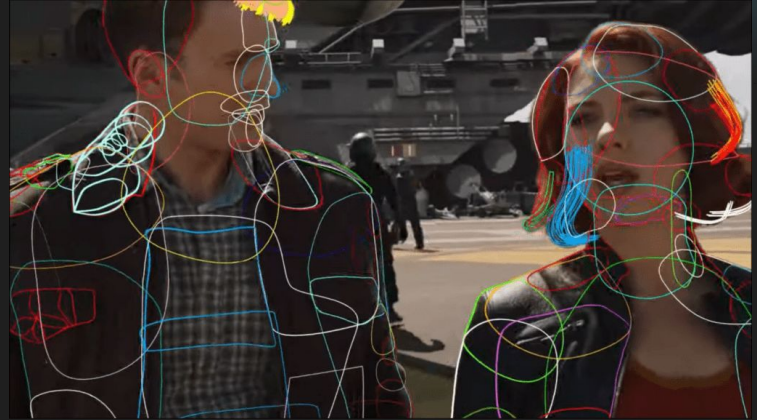
Current video editing workflows require tedious, frame-by-frame manual object selection to remove unwanted elements.

We propose an end-to-end agentic pipeline that allows users to erase objects from videos using a single natural language prompt (e.g., "Remove the dog on the left").

Our system integrates GroundingDINO and SAM2 for precise text-to-mask grounding, and leverages an adaptive "Prompt Complexity Router." This novel orchestrator dynamically selects the optimal inpainting backend (e.g., LaMa, ProPainter or E2FGVI) based on text complexity and motion, balancing inference latency with high-fidelity, temporally consistent video reconstruction.

Problem Statement

- **Manual Bottleneck:** Removing objects in standard video editors (like Premiere Pro or After Effects) requires manual rotoscoping or precise masking across hundreds of frames.
- **Static vs. Dynamic:** While image inpainting (Magic Eraser) is solved, video inpainting suffers from temporal flickering, tracking drift, and high computational costs.



Manual rotoscoping in After Effects involves precise masking across hundreds of frames

Relevant Research

01

Text Guided Detection

Slide 5

02

Video Object Tracking and Propagation

Slide 6

03

Video Inpainting and Completion

Slide 7

04

Language Guided Video Editing

Slide 8

05

Agentic / Routing System

Slide 9

Research Area 1: Text Guided Detection

Paper Title	Core innovation	Why we need it
GroundingDINO : Marrying DINO with Grounded Pre-Training for Open-Set Object Detection (2023)	→ Zero-shot text-to-bbox detection using language grounding → Achieves 52.5 AP on COCO without training data	→ Foundation for first-frame object detection from text → Handles novel object categories via open-vocabulary
SAM 2 : Segment Anything in Images and Videos (2024)	→ Promptable video segmentation with memory module → 6× faster than SAM with improved accuracy	→ Converts bbox to precise masks across video frames → Memory enables tracking through occlusions
OWL-ViT : Simple Open-Vocabulary Object Detection with Vision Transformers (2022)	→ First open-vocabulary detector for zero-shot text-to-object detection → Scales to arbitrary text queries without retraining (e.g., "red sports car")	→ Backup detector for complex prompts → Zero-shot capability ensures pipeline works on unseen categories without dataset collection
LoVoRA : Text-guided and Mask-free Video Object Removal via Large Multimodal Models (Dec 2025)	→ End-to-end text-guided video removal (GroundingDINO + SAM2) → SOTA on ROSE benchmark with side-effect removal	→ Inspires adaptive router design (confidence-based propagation) → Direct blueprint for our pipeline architecture (text → detect → track → inpaint)
CLIPSeg : Image Segmentation Using Text and Image Prompts (ECCV 2022)	→ CLIP-based text-to-segmentation decoder → Zero-shot segmentation from language prompts	→ Semantic guidance for coarse mask generation → Lightweight alternative for frame-level seg⁶

Research Area 2: Video Object Tracking and Propagation

Paper Title	Core innovation	Why we need it
RAFT : Recurrent All-Pairs Field Transforms for Optical Flow (ECCV 2020)	→ Iterative refinement for optical flow estimation → SOTA on Sintel/KITTI with high efficiency	→ Mask warping across frames via flow fields → Maintains temporal coherence in propagation
XMem : Long-Term Video Object Segmentation with Atkinson-Shiffrin Memory (ECCV 2022)	→ Memory transformer for long-range tracking (J&F 87.7) → Unified sensory/working/long-term memory stores	→ Handles occlusions in extended videos → Tracks objects through disappearance/reappearance
Cutie : Putting the Object Back into Video Object Segmentation (CVPR 2024)	→ Object-level memory reading (J&F 91.5 on DAVIS) → Query-based transformer for distractor separation	→ Most accurate tracking with efficient memory → Robust to similar objects (distractor handling)
VideoSAM : SAM Adaptation for High-Speed Video Segmentation (2024)	→ Fine-tuned SAM for temporal video segmentation → Superior performance on diverse fluid environments	→ Extends SAM's promptability to video domain → Lightweight alternative to heavy trackers

Research Area 3: Video Inpainting and Completion

Paper Title	Core innovation	Why we need it
Deep Flow-Guided Video Inpainting (CVPR 2019)	→ First deep flow-based inpainting framework → Completes flow → propagates pixels for coherence	→ Pioneering baseline for flow-guided methods → Establishes temporal consistency paradigm
E²FGVI : End-to-End Framework for Flow-Guided Video Inpainting (CVPR 2022)	→ End-to-end trainable flow completion + propagation → Achieves efficiency with joint optimization	→ Primary fast inpainting backend (0.5s/frame) → Balanced quality-speed for router selection
ProPainter : Improving Propagation and Transformer for Video Inpainting (ICCV 2023)	→ Dual-domain propagation (image + feature warping) → Mask-guided sparse transformer (1.46 dB PSNR gain)	→ High-quality backend for complex scenes → Router selects when accuracy prioritized
ROSE : Remove Objects with Side Effects in Videos (Aug 2025)	→ Explicit side-effect prediction (shadows/reflections) → ROSE-Bench for comprehensive evaluation	→ Handles environmental effects (not just object) → Benchmark for router performance evaluation
VideoPainter : Any-length Video Inpainting with Context Control (SIGGRAPH 2025)	→ Dual-branch framework with lightweight encoder (6% params) → ID-consistent editing for arbitrary length	→ Scalable architecture for long video processing → Plug-and-play stylization inspiration

Research Area 4: Language Guided Video Editing

Paper Title	Core innovation	Why we need it
LDVI: Language-Driven Video Inpainting via Multimodal LLMs (2024)	→ Natural language + conversation-based inpainting → MLLM-enhanced interactive architecture (LGVI-I)	→ End-to-end language grounding for inpainting → Inspires conversational UI design
Text-Guided Video Amodal Completion (ICCV 2025)	→ Diffusion-based text-guided amodal object completion → Zero-shot inference for long videos	→ Handles occluded objects via text guidance → Motion continuity for temporal consistency

Research Area 5: Agentic / Routing System

Paper Title	Core innovation	Why we need it
Picsellia Atlas: Solving Computer Vision with AI Agents (2025)	→ Multi-agent CV workflow → SmolAgents orchestration with Pydantic-AI validation	→ Blueprint for agent architecture design → Router + orchestration framework inspiration
CrewAI: Open Source Multi-Agent Orchestration Framework (2024)	→ Specialized planning/reasoning agents with collaboration → Context sharing and delegation for complex tasks	→ Framework for implementing adaptive router → Multi-agent task coordination patterns
OmniEraser: Remove Objects and Effects in Images with Diffusion (2025)	→ Video4Removal dataset (100K+ image pairs from video) → Object-Background Guidance to prevent artifacts	→ Side-effect removal methodology → Dataset construction approach from video frames
VORNet: Spatio-Temporally Consistent Video Inpainting for Object Removal (CVPR 2019)	→ 3D CNN for spatial-temporal feature learning → Copy-and-paste network for pixel synthesis	→ Early work on end-to-end video object removal → Establishes consistency evaluation metrics
EraserDiT: Fast Video Inpainting with Diffusion Transformer Model (2025)	→ Reducing inference latency for high-resolution video inpainting without dropping temporal consistency	→ “Fast” vs. “Slow” routing → High processing speed for high resolution high frame rate videos

Proposed Solution

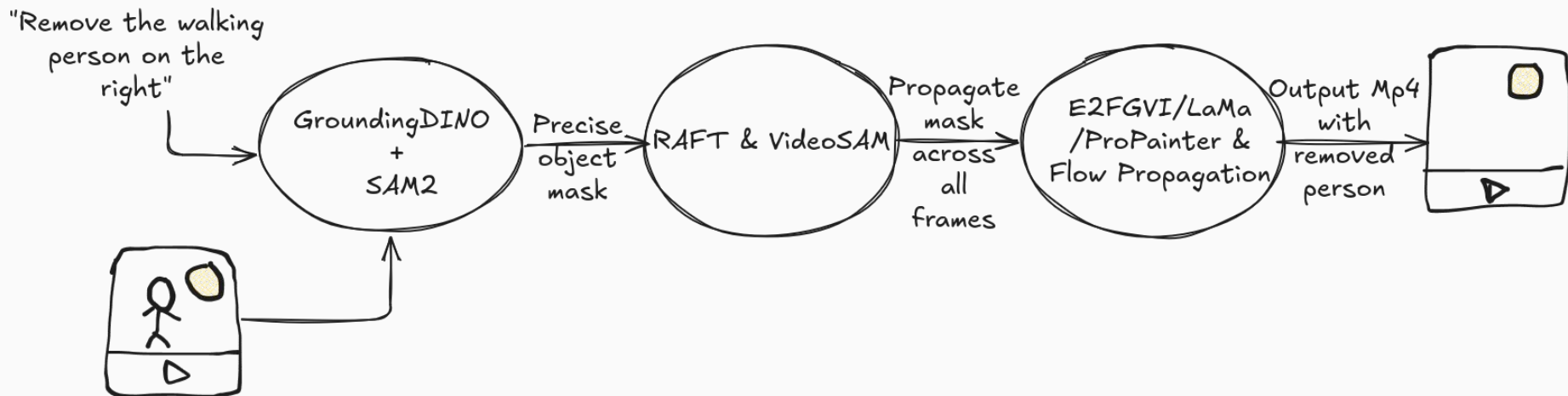
- **Zero-Shot NLP Interface:** Users provide a raw video and a simple text prompt. No manual drawing is required.
- **Automated Pipeline:**
 - a. **Grounding:** Text translates to a precise object mask in the first frame.
 - b. **Propagation:** The mask tracks the object across all frames, even through motion.
 - c. **Erase:** The masked area is contextually in-painted using temporal data from surrounding frames to ensure the background looks natural.
- **Agentic Approach:** The system monitors its own success (confidence scores) and dynamically routes tasks to the most efficient model.

System Architecture



- **Text to object detection:** Take the text prompt and frame 1 of the video as input. GroundingDino outputs the bounding box and SAM2 will generate a pixel perfect mask.
- **Temporal tracking:** We need to propagate SAM2's mask across all frames in the video. RAFT/VideoSAM will be used for this.
- **Adaptive Orchestration (Our Core Contribution):** Different models work well for different kinds of objects and videos. The task here is to analyze the prompt and video to route to the best inpainting backend. The inpainting backend will fill the masked area temporally across the frames.
- **Inpainting and reconstruction:** Fill the masked area. Models: E2FGVI (flow-specialized), ProPainter (complex/diffusion), LaMa (simple/fast).

System Architecture



Novelty and Our Contribution

1



Prompt complexity router:

This layer will extract features like prompt length, number of objects and motion hints.

These features will help us route tasks to the different inpainting backend we have.

For example, simple tasks have speed as priority and can be routed to LaMa, while complex tasks can be routed to high quality models like ProPainter or E2FGVI

2



Keyframe Propagation with Error Correction:

To prevent tracking drift, the system will query GroundingDino every 10th frame.

This will ensure that we are in a position to handle heavy occlusions (culling some important features due to mask drift) or temporary tracking failures.

Regenerating the bounding box and mask helps it stay focused on the object.

3



Temporal consistency score:

We calculate the warping errors between each frame to generate a conscience score.

Low confidence means the inpainting is not well done and we need to maintain better quality of inpainting. So, we may fall back on a more robust inpainting backend.

Novelty Technical Details

1



Prompt complexity router:

$$y_{\text{pred}} = \text{Softmax}(\text{MLP}(f_{\text{text}}, N_{\text{obj}}, \mu_{\text{motion}}))$$

where,

f_{text} is the text features,

N_{obj} is the object count, and

M_{motion} is the motion variance

2



Anti-Flicker Blending (Exponential Moving Average)

To prevent high frequency flicker in the final output, we apply EMA blending.

$$V_t = \alpha V_t + (1 - \alpha) W(V_{t-1}, f_{t \rightarrow t-1})$$

where,

V_t is the output,

α is blending decay rate, and

F is optical flow.

3



Temporal consistency score:

$$E_{\text{warp}} = 1/N * \sum ||I_t - W(I_{t-1}, F_{t \rightarrow t-1})||_1 \odot M_t$$

where,

W is backward warping operation,

F is optical flow, and

M_t is inpainting mask

Week 0

Work done until first presentation



Literature survey and finalizing the architecture

Surveyed 21 papers and came to a consensus regarding what the final deliverables are going to be.

Brainstormed ideas to come up with a system architecture that incorporates the learnings from the papers.

Played around with a few inpainting models like LaMa to understand how the frontend works and had a brief look at the architecture and code.

Project Timelines

Week 1



Baseline & Grounding Setup

Set up virtual environments and download pre-trained weights.
Pipeline integration: Connect GroundingDINO to SAM2.
Text prompt successfully outputs a precise segmentation mask on a single frame.

Week 2



Tracking & Flow Propagation

Integrate RAFT (optical flow) and VideoSAM.
Implement basic mask propagation across a 30-frame video clip.
Mask successfully follows the object throughout a short video.

Week 3



Inpainting Integration & UI

Connect the 3 inpainting backends: LaMa (image-based baseline), E2FGVI, and ProPainter.
Build the basic Gradio User Interface.

Project Timelines

Week 4



Implementation of MLP, error correction and temporal consistency

Build and train the Prompt Complexity Router.
Implement the 10-frame Error Correction re-query loop.
Implement the Temporal Consistency Score and fallback logic.

Week 5



Evaluation, Ablation & Final Delivery

Run benchmarks: Temporal LPIPS (target < 0.15) and Tracking IoU (target > 0.85).
Conduct ablation studies (e.g., Router vs. No Router latency comparison).
Finalize technical report, record the 30s demo video, and clean the GitHub repo

Week 6



Buffer for unforeseen delays

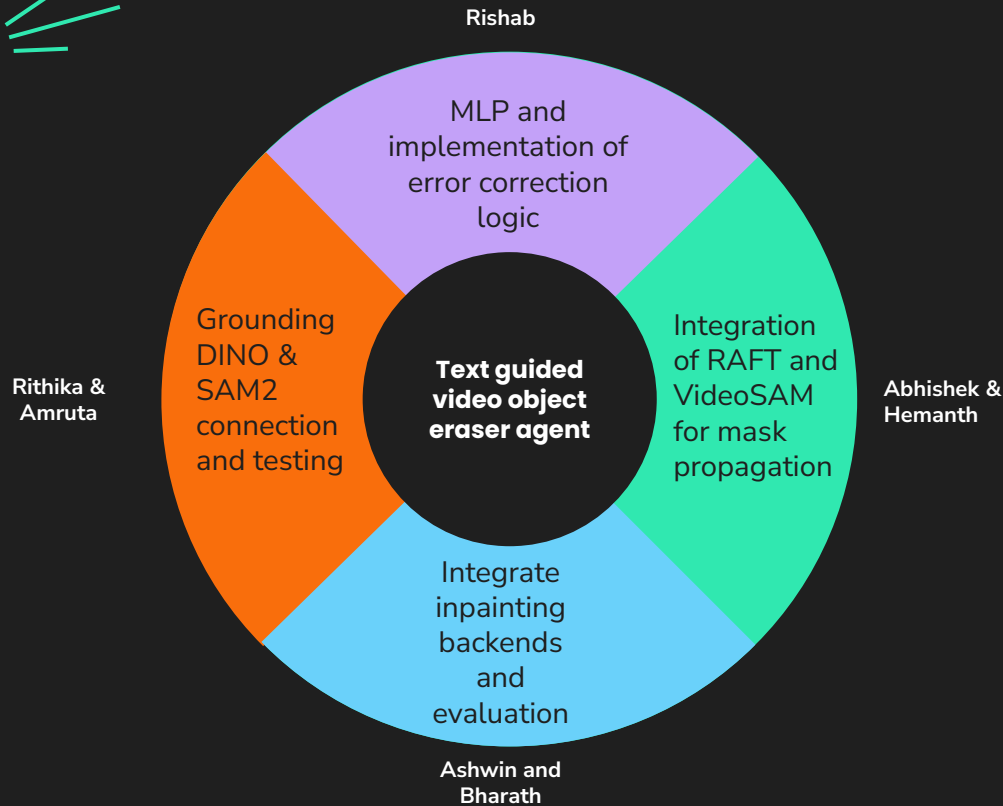
Complete report if not done.
Code cleanup and resolution of issues.



Final Deliverables & Future Improvements

- Deliverables: A working Gradio web prototype and a complete technical report detailing our project.
- Future improvements: Real-Time Processing by further optimizing the pipeline with TensorRT for live-stream object removal.
- Improving the memory bank for when an object completely leaves the frame and returns.

Team Contributions



Thank you!
